TalentTrace AI — Technical System Design

React / FastAPI modular monolith · LangGraph-assisted workflows · evidence-backed hiring

A · PRESENTATION & ACCESS

React + Vite + TypeScript

- **JD / Publishing** — Save drafts · review channels
- **Review Applicants** — Criteria · CVs · ranked results
- **Interview & Scheduling** — Recruiter portal · candidate links
- **Assessment & F11** — Reports · recruiter-only prompts

HTTPS REST / authenticated session · event updates

B · FASTAPI BACKEND / MODULAR MONOLITH

API routing · recruiter/job ownership · consent gates · business rules · versioned records

CORE APPLICATION SERVICES

- **JD & policy services** — Draft/refine revisions; Review/approve criteria
- **Publication + intake** — Discord MCP / Telegram Bot; Google Forms / Drive bridge
- **CV screening F1/F2** — PDF/DOCX · Tesseract OCR; Weights + saved evidence
- **Interview F3/F4** — Booking · LiveKit rooms; AssemblyAI + transcript
- **Evaluation F5/F6** — Technical RAG · Q&A scoring; Report + comparison

LANGGRAPH-ASSISTED WORKFLOWS

bounded state · no unnecessary master graph

- **JD assistant** — Grounding + revision
- **CV screening** — Evidence + routing
- **F11 live co-pilot** — Private suggestion stream
- **F5 Q&A analysis** — Grounded evaluation

SHARED AI & SAFETY

- **LLM provider adapter** — Groq API primary · optional local model path
- **RAG knowledge services** — Company policy / technical references
- **Security and consent boundary** — Candidate isolation · external processing consent

C · EXTERNAL ADAPTERS

Explicit, scoped provider connections

- **Groq API** — JD / CV / supported evaluation calls
- **Discord via MCP** — Explicit approval · receipt
- **Telegram Bot API** — Explicit approval · receipt
- **Google Forms + Drive** — Scoped CV import · one-shot operator
- **LiveKit** — WebRTC two-person media
- **AssemblyAI** — Speaker-attributed live STT path
- **SMTP / Email** — Configured interview notifications

D · PERSISTENCE, FILES & AUDITABLE EVIDENCE

- **PostgreSQL business records** — Recruiter · jobs · criteria revisions; CV submissions · screening scores
- **Interview & assessment data** — Slots · invitations · transcript turns; F5/F6 results · comparison
- **LangGraph + vectors** — Workflow checkpoints / state; pgvector & RAG indexes
- **Private CV file storage** — PDF/DOCX in protected volume; No public candidate file URLs

Speaker transcript storage is consent-controlled. Raw interview audio/video is not intended for permanent storage.

DATA FLOW

Job revision → publication → controlled CV intake → screening → interview + captions → F5 / F6 report → human decision